

# BoundaryLens: SIH Presentation Speaking Guide

## 3D ULPIN-Linked Vertical Property Delineation Prototype • Presenter Script & Cheat Sheet

**Team:**  
areonerds

**Pilot Region:** Bengaluru Urban  
(Koramangala/BTM)

**Status:** DEMO READY (124/124 Tests Passed)

### CORE ETHICAL & LEGAL DIRECTIVE (AGENTS.md):

Never claim that BoundaryLens officially generates or issues legal government ULPINs. BoundaryLens creates "**proposed 3D vertical spatial linkages**" for administrative evaluation and reviewer audit. Official legal title issuance remains the sole jurisdiction of competent revenue authorities.

## Part 1: Slide-by-Slide Speaking Script

This script is formatted for a 14-slide deck. Each slide contains the exact words to speak, key bullet points for the slide, technical terms explained simply, speaking duration, and anticipated judge questions.

## Slide 1: Title & Team Introduction

🕒 25 Seconds

### WHAT IS ON PPT:

- **BoundaryLens**: 3D ULPIN-Linked Vertical Property Delineation Prototype
- Smart India Hackathon 2026 | Problem Statement: SIH26011
- Team: **areonerds** | Pilot Area: Bengaluru Urban (Koramangala / BTM)

### 🗣️ EXACTLY WHAT TO SAY:

"Good morning, respected judges. We are **Team areonerds**, and today we present **BoundaryLens** — an AI and GIS powered prototype for **3D ULPIN-linked vertical property mapping** under problem statement **SIH26011**.

As India's cities expand upwards with high-rises and multi-storey apartments, our current land records remain locked in flat, two-dimensional maps. Today, we demonstrate how open geospatial datasets, elevation models, and machine learning can deterministically connect 2D surface parcels to 3D vertical units — without fabricating government records."

### TERM EXPLAINED SIMPLY:

**ULPIN**: Unique Land Parcel Identification Number — think of it as an 'Aadhaar Card' for a plot of land, currently only existing in 2D.

### KEY POINT TO EMPHASIZE:

Introduce the team confidently; state clearly that we solve vertical property mapping without fake data.

### Q: Why did you name it **BoundaryLens**?

A: Because it acts as an optical lens on cadastral boundaries, allowing revenue officers to zoom from a 2D surface parcel directly into vertical floor volumes and spatial relationships.

## Slide 2: Problem Statement (SIH26011)

🕒 30 Seconds

### WHAT IS ON PPT:

- SIH26011: Assigning 3D identities for surface parcels, multi-storey properties & underground infrastructure.
- 2D land records represent only surface boundaries (X, Y).
- Modern urban reality: Multi-tier vertical ownership, apartments, commercial floors (Z-axis).

Transition: "Now, let us examine the core problem statement..."

### 🔑 EXACTLY WHAT TO SAY:

"Problem Statement SIH26011 asks for **3D identities** for surface land, multi-storey properties, and underground utilities.

Today, if a 10-storey apartment building sits on a single land parcel, government cadastral records only register the single 2D ground boundary. There is no official spatial identity connecting an individual 4th-floor flat to the revenue land parcel beneath it.

This leads to property disputes, tax leakage, overlapping claims, and makes 3D city planning impossible."

### TERM EXPLAINED SIMPLY:

**Cadastral Map:** An official government survey map showing legal boundaries, plot numbers, and ownership borders.

### KEY POINT TO EMPHASIZE:

India has 2D digitisation progress, but zero spatial mechanism for vertical property stacks.

### Q: Doesn't the municipality already have floor plans?

A: Municipalities have architectural floor plans as static PDFs or blueprints, but they are not topologically linked to the GIS cadastral parcel system.

### Slide 3: Why This Problem is Difficult

🕒 30 Seconds

#### WHAT IS ON PPT:

- Disparate Data Silos: Cadastral revenue maps vs OpenStreetMap vs Satellite imagery.
- Resolution Mismatch: 30-metre global satellite elevation vs 5-metre urban buildings.
- The "Hallucination Trap": Many prototypes invent fake heights or fabricate ULPINs.

Transition: "Why hasn't this been solved already? There are three huge challenges..."

#### 🔪 EXACTLY WHAT TO SAY:

"First, **data sources disagree**. Land registry maps, satellite imagery, and municipal layers operate in different coordinate systems with misalignments.

Second, **public satellite elevation data is coarse** — 30-metre pixels cannot separate individual floor slabs directly.

And third, the biggest risk in hackathons is **fabricated data**: prototypes that divide elevation by 3.5 metres and pretend they have exact architectural blueprints. In property law, guessing is dangerous. If evidence is missing, a system must honestly say *NOT DETERMINABLE*."

#### TERM EXPLAINED SIMPLY:

**Coarse Resolution:** Like a blurry camera where 1 pixel covers 30 meters on the ground; fine architectural details cannot be seen directly.

#### KEY POINT TO EMPHASIZE:

Highlight our integrity: we refuse to fabricate exact heights where coarse data cannot support it.

#### Q: Why can't you just divide building height by 3.5m to get floors?

A: Because a 12m structure could be a 3-storey apartment, a high-ceiling commercial warehouse, or tree canopy covering a small roof. Real floor counts require supervised ML and evidence fusion.

## Slide 4: Our Solution — BoundaryLens

🕒 35 Seconds

### WHAT IS ON PPT:

- **Deterministic GIS matching:** Links 2D parcel polygons to building footprints.
- **Dual-Raster Elevation:** Bare-earth DEM (terrain ground) + GLO-30 DSM (surface).
- **Supervised ML Floor Estimation:** Random Forest trained on 12,700+ real OSM labels.
- **AI Anomaly Detection:** Isolation Forest flags boundary conflicts for human audit.
- **Proposed 3D Vertical Spatial Linkage:** Structure: IN-KA-BLR-P<Parcel>-B<Bldg>-F<Floor>.

Transition: "This brings us to our solution: BoundaryLens..."

### 🔑 EXACTLY WHAT TO SAY:

"BoundaryLens is a complete, modular pipeline and 3D Web UI.

We start with **2D deterministic topology** to check whether a building cleanly sits inside its legal parcel. Next, we sample **bare-earth DEM** to establish true ground elevation.

Then, we fuse satellite vegetation evidence and machine learning to estimate vertical floors, allocate proposed 3D identifiers, and flag boundary encroachments for human registrar review. Everything is traceable back to an immutable evidence ledger."

### TERM EXPLAINED SIMPLY:

**Topology:** Mathematical geometry rules that test whether shapes touch, intersect, or cleanly sit inside one another.

### KEY POINT TO EMPHASIZE:

Show the complete hierarchy: Parcel (2D) → Building (Mass) → Floor (Volume) → Unit Identity.

### Q: What is the format of your proposed 3D ID?

A: IN-KA-BLR-P22068-B93697573-F2 represents: Country (IN), State (KA), District (BLR), Revenue Parcel (22068), Building Footprint, and Floor 2.

## Slide 5: End-to-End System Workflow

🕒 30 Seconds

### WHAT IS ON PPT:

- Phase 1–2: Ingest OpenCity Cadastral, OSM Footprints, Copernicus DSM, Bare-Earth DEM.
- Phase 3–5: Normalize CRS (EPSG:4326/32643) & Topological 2D Intersection Matching.
- Phase 6–10: Elevation Extraction, NDVI Sentinel-2 Evidence & AI Anomaly Isolation Forest.
- Phase 11–15: Floor ML Modeling, Evidence Fusion Ledger & MapLibre 3D Web Interface.

Transition: "Here is how data flows through our 17-phase architecture..."

### 🔪 EXACTLY WHAT TO SAY:

"Our system follows a strict 17-phase progression governed by our project constitution `AGENTS.md`.

Raw layers are normalized into standard coordinate systems. The 2D matching engine categorizes each building as **CONTAINED**, **MAJORITY**, or **CONFLICT**.

Elevation and vegetation models extract physical signals, our AI Isolation Forest checks for multi-dimensional anomalies, and the fused result is staged directly into our hardware-accelerated 3D dashboard."

### TERM EXPLAINED SIMPLY:

**CRS (Coordinate Reference System):** The mathematical grid used to place satellite coordinates onto a curved Earth accurately.

### KEY POINT TO EMPHASIZE:

Modular architecture: Every phase writes standalone verifiable GeoJSON/JSON artifacts.

### Q: What happens if a phase fails?

A: Under our constitution, if an essential layer fails or CRS is incompatible, execution halts rather than fabricating a default fallback.

## Slide 6: Multi-Tiered Open Geospatial Data

🕒 30 Seconds

### WHAT IS ON PPT:

- **Cadastral Parcels:** OpenCity Bengaluru Municipal Revenue Layer (KMZ/GeoJSON).
- **Building Footprints:** OpenStreetMap vector polygons (2,734 AOI buildings).
- **Terrain Elevation (DEM):** Bare-earth SRTM 30m / AWS Terrain (Z-axis ground level).
- **Surface Elevation (DSM):** Copernicus GLO-30 Digital Surface Model (~30m).
- **Vegetation Evidence:** Sentinel-2 L2A 10m Optical Bands (NDVI Canopy Mask).

Transition: "Let us examine the datasets powering this pilot..."

### 🔑 EXACTLY WHAT TO SAY:

"For our pilot area in South Bengaluru, we use **100% verified, legal open datasets**.

Cadastral boundaries come from the OpenCity municipal GIS repository. Footprints come from OpenStreetMap. Ground elevation comes from a bare-earth DEM, surface elevation comes from Copernicus GLO-30, and canopy signals come from Sentinel-2 satellite imagery.

Every dataset has a signed manifest in our data registry detailing provenance, acquisition date, resolution, and licensing."

### TERM EXPLAINED SIMPLY:

**DEM vs DSM:** DEM is bare ground (dirt/rocks); DSM is surface (including buildings and tree tops).

### KEY POINT TO EMPHASIZE:

Tiered Data Strategy: No synthetic data is ever called government data.

### Q: Why choose South Bengaluru for the pilot?

A: Because it possesses high-density multi-storey properties alongside published digitised cadastral boundary maps, providing an ideal stress test.

## Slide 7: 2D Spatial Matching & Ground Elevation

🕒 30 Seconds

### WHAT IS ON PPT:

- Topological Intersection: Polygon-in-polygon math using Shapely and GeoPandas.
- Overlap Ratio Classification:
  - >95% Overlap → **CONTAINED** (● Green)
  - 50%–95% Overlap → **MAJORITY** (● Amber)
  - <50% or multi-parcel crossing → **CONFLICT** (● Red)
- Ground Elevation: Sampled at building footprint centroid from bare-earth DEM (e.g. 904.19 m).

Transition: "Now, how do we link a building to a land parcel? We use deterministic GIS..."

### 🔑 EXACTLY WHAT TO SAY:

"Before touching 3D, we must master 2D topology.

Using Shapely geometry operations, we calculate the exact spatial intersection between building footprints and legal parcel boundaries. If a building is cleanly inside, it is marked **CONTAINED** in green.

At the same time, we sample the bare-earth DEM at the building's centroid to extract its true elevation above mean sea level — establishing the exact Z-axis foundation for vertical mapping."

### TERM EXPLAINED SIMPLY:

**Centroid:** The exact geographic center point of a building footprint polygon.

### KEY POINT TO EMPHASIZE:

Deterministic logic: Never replace exact geometry operations with an opaque machine learning guess.

### Q: What happens if a building crosses two parcels?

A: It is flagged with status `BOUNDARY\_OVERLAP` / `CONFLICT` in red and routed directly to the human reviewer audit queue.

## Slide 8: 3D Height Derivation & Evidence Hierarchy

🕒 35 Seconds

### WHAT IS ON PPT:

- Evidence Hierarchy (Rule 3): Never silently pick a source when sources disagree.
- Source 1: **OSM Verified Levels** → When explicit tag exists, status is OSM\_VERIFIED.
- Source 2: **Urban Profile Model** → When unobserved, height is ESTIMATED / PROVISIONAL.
- DSM - DEM Signal: Coarse elevation delta used as supporting feature, not exact floor count.

Transition: "Moving from 2D to 3D requires strict evidence hierarchy..."

### 🔑 EXACTLY WHAT TO SAY:

"In BoundaryLens, every data field carries its source and confidence status.

When an official OSM surveyor tag exists, we treat it as **OBSERVED** evidence. But when levels are unobserved, we do not pretend we measured it with a laser. We clearly tag it as **ESTIMATED (Urban Profile Model)** with provisional status.

Furthermore, we calculate the delta between Copernicus DSM surface height and terrain DEM to provide a coarse physical height signal as secondary validation."

### TERM EXPLAINED SIMPLY:

**Evidence Hierarchy:** A strict ranking rule:  
Verified ground survey > Model prediction >  
Coarse satellite signal.

### KEY POINT TO EMPHASIZE:

Honesty in reporting: We never present an estimated height as a legally measured reality.

### Q: Why not use drone LiDAR for all buildings?

A: In production, governments will integrate drone LiDAR where available. But BoundaryLens proves how far open satellite data can take you right now before drone flights occur.

## Slide 9: Calibrated ML Floor Estimation (Phase 12)

🕒 35 Seconds

### WHAT IS ON PPT:

- Training Data: **12,762 real OSM building:levels labels** across greater Bengaluru.
- Spatial Split: 182 spatial grid cells (Train: 8,461 | Val: 2,329 | Test: 1,972) — zero spatial leakage.
- Model: **Random Forest Regressor** | Test MAE: **2.22 floors** |  $\pm 1$  Accuracy: **59.8%**.
- Independent AOI Check (n=16): MAE **1.21 floors** |  $\pm 1$  Accuracy: **62.5%**.
- Three Clear States: OBSERVED (16) | PREDICTED (2,453) | NOT\_DETERMINABLE (265).

Transition: "Let's look at our machine learning floor model..."

### 🔑 EXACTLY WHAT TO SAY:

"Instead of applying a blind mathematical division, we trained a **supervised Random Forest model** on 12,762 real Bengaluru building levels.

To prevent data leakage, we split our data across 182 spatial grid cells. On the test set, our model achieves a Mean Absolute Error of 2.22 floors and accurately predicts within  $\pm 1$  floor roughly 60% of the time.

Most importantly, if footprint size or elevation variance is ambiguous, the model abstains and outputs **NOT DETERMINABLE**. We never guess."

### TERM EXPLAINED SIMPLY:

**Spatial Split:** Training on certain neighborhoods and testing on completely separate neighborhoods so the model doesn't cheat by memorizing nearby houses.

### KEY POINT TO EMPHASIZE:

Real metrics: 12,762 real labels, 59.8%  $\pm 1$  accuracy, calibrated confidence.

### Q: Why is accuracy 60% and not 95%?

A: Because 60% is an honest, real-world metric on 30m open satellite data without high-resolution drone imagery. Anyone claiming 98% accuracy without drone LiDAR is overfitting or using synthetic data.

## Slide 10: AI Anomaly Detection & Evidence Fusion

🕒 30 Seconds

### WHAT IS ON PPT:

- **Unsupervised Isolation Forest:** Evaluates footprint area, perimeter, parcel overlap ratio, elevation.
- Flags irregular geometries, severe encroachments, and out-of-distribution building forms.
- **Deterministic Evidence Fusion Engine:** Compiles multi-layer data into an immutable JSON ledger.
- AI assists; it does *not* adjudicate ownership or legal disputes.

Transition: "To catch boundary conflicts automatically, we use AI anomaly detection..."

### 🔑 EXACTLY WHAT TO SAY:

"We implement an **Isolation Forest** from scikit-learn to spot geometric and topological anomalies.

The model flags buildings that cross boundary lines, possess abnormal footprint-to-parcel ratios, or exhibit strange height profiles. These structures are flagged with negative anomaly scores.

Our Evidence Fusion Engine logs every feature into an immutable ledger, automatically routing anomalous structures to our human review gate."

### TERM EXPLAINED SIMPLY:

**Isolation Forest:** An AI algorithm that detects strange or unusual data points by seeing how easily they can be isolated from normal data patterns.

### KEY POINT TO EMPHASIZE:

AI serves as a flagger and decision-support assistant, never a legal judge.

### Q: What triggers an anomaly flag?

A: Significant boundary encroachment, extreme footprint shape distortion, or severe height-to-area inconsistency.

## Slide 11: Interactive MapLibre 3D Dashboard

🕒 35 Seconds

### WHAT IS ON PPT:

- Frontend Stack: Pure HTML5, CSS3, JavaScript + **MapLibre GL JS** (hardware-accelerated WebGL).
- Zero build step, dependency-light, runs 100% locally and offline on port 8000.
- Features: 3D extruded polygons, revenue parcel overlays, property inspector, NDVI vegetation toggle.
- Integrated **Three.js Floor Visualizer**: Renders stacked 3D floor slabs and floor spatial IDs.

Transition: "Now, let us look at the visual interface our surveyors use..."

### 🔑 EXACTLY WHAT TO SAY:

"Our frontend is built with **MapLibre GL JS** and vanilla JavaScript. We deliberately avoided heavy frontend frameworks so the entire 3D interface loads in milliseconds without internet dependencies.

Surveyors can tilt, rotate, and inspect 2,734 building volumes in 3D. Clicking any structure opens the Property Card displaying parcel linkage, ground elevation, floor breakdowns, and AI anomaly scores.

When a building has floor estimates, clicking 'View Floor Levels' renders discrete 3D floor slabs."

### TERM EXPLAINED SIMPLY:

**MapLibre GL**: A high-performance open-source 3D mapping engine that uses your computer's graphics card (GPU) to render thousands of buildings smoothly.

### KEY POINT TO EMPHASIZE:

Reliable & fast: Runs offline on localhost:8000; zero risk of loading spinners or broken build steps.

### Q: Can this run in a government office with slow internet?

A: Yes! Once cached or running on an intranet GIS server, MapLibre requires zero external API roundtrips.

## Slide 12: Human Verification & Reviewer Audit Gate

🕒 35 Seconds

### WHAT IS ON PPT:

- Rule 8 Compliance: Legally significant uncertainty is surfaced to a human registrar.
- Reviewer Actions: APPROVE (Green) | CORRECT (Blue) | REJECT (Red) | UNRESOLVED (Amber).
- Live Activity Log: Every reviewer decision is captured with timestamp, action type, and parcel reference.
- Audit Trail: Session activity stream guarantees accountability before any vertical identity is endorsed.

Transition: "This brings us to the most important governance feature: Human Verification..."

### 🔑 EXACTLY WHAT TO SAY:

"In property governance, an AI algorithm must never make final legal decisions. Under Rule 8 of our constitution, whenever an anomaly is detected, the Verification Gate changes to **HUMAN VERIFICATION REQUIRED**.

A municipal officer reviews the evidence and can click **APPROVE, CORRECT, REJECT, or MARK UNRESOLVED**.

The building updates its visual color immediately, and the action is recorded in the **Registrar Activity Log** with an immutable timestamp for full legal auditability."

### TERM EXPLAINED SIMPLY:

**Human-in-the-Loop:** An engineering design where AI flags problems, but an authorized human officer makes the final binding decision.

### KEY POINT TO EMPHASIZE:

Judges love legal accountability: Show how our audit gate prevents automated legal blunders.

### Q: What is the difference between REJECT and UNRESOLVED?

A: `REJECT` denies an invalid boundary alignment outright; `UNRESOLVED` flags the property for an on-ground physical total station survey.

## Slide 13: Results, Validation & Benchmarks

🕒 30 Seconds

### WHAT IS ON PPT:

- Automated Test Suite: **124 of 124 unit & contract tests PASSED (100%)** in 1.70s.
- Pilot Scale: **2,734 buildings** mapped across **26 cadastral parcels**.
- Floor Detection: 16 observed tags, 2,453 calibrated predictions, 265 non-determinable.
- Elevation Benchmarks: Min terrain 891.4 m, Max 918.7 m, Mean 904.2 m (100% DEM coverage).
- Demo Health Check: 7/7 core subsystem tests passing; zero runtime errors.

Transition: "Here are our verified empirical test results..."

### 🔑 EXACTLY WHAT TO SAY:

"We have rigorously validated our prototype with **124 automated unit and contract tests** covering geometry validity, DEM sampling, NDVI formulas, and frontend API contracts — with a 100% pass rate.

Across our Bengaluru pilot, we processed 2,734 structures. 2,453 received calibrated floor predictions, 16 carry ground-truth observed levels, and 265 were responsibly marked non-determinable.

Our pre-flight demo health check confirms all data layers and local endpoints are active."

### TERM EXPLAINED SIMPLY:

**Contract Test:** A test that guarantees the backend data format matches exactly what the frontend user interface expects to receive.

### KEY POINT TO EMPHASIZE:

124/124 tests passing proves software maturity, not just a hackathon wireframe.

### Q: Have you tested this outside Bengaluru?

A: Yes, our pipeline has test configurations for Chennai DRDO and Baramati ULB; the modular pipeline accepts any standard GeoJSON/raster pair.

## Slide 14: Live Demonstration Protocol

🕒 45 Seconds

### WHAT IS ON PPT:

- Case 1: **Normal Building** (`osm\_way\_93697573`) → Contained (●), 904.19m ground elevation, proposed 3D linkage.
- Case 2: **Boundary Conflict** (`osm\_way\_130829155`) → Anomaly detected, routed to human verification.
- Case 3: **Multi-Storey Floor Stack** (`osm\_way\_347151672`) → Real floor levels, discrete floor spatial IDs.
- Interactive Reviewer Action: Click APPROVE/REJECT → View live Registrar Activity Log.

Transition: "Now, let us show you BoundaryLens in live action..."

### 🔊 EXACTLY WHAT TO SAY:

"Let us switch to our live screen on <http://localhost:8000>.

First, I click this green building: notice it sits cleanly inside Cadastral Parcel 22068 with 904.2m ground elevation. Its proposed vertical linkage is IN-KA-BLR-P22068-B93697573.

Second, I click this anomalous building: notice our Isolation Forest flagged it with an anomaly score of -0.036. The gate demands **HUMAN VERIFICATION REQUIRED**.

As the registrar, I review it and click **REJECT**. The building instantly updates on the 3D map, and opening our **Audit Logs** tab proves the decision was permanently logged."

### DEMO TIP:

Keep the browser zoomed to 100%; click deliberately; tilt the map with right-click drag to show true 3D depth.

### KEY POINT TO EMPHASIZE:

Live interactive proof of AI detection → Reviewer action → Audit log update.

### Q: What is the purpose of the 1m Drone toggle?

A: It demonstrates how our UI will behave when municipal authorities supply 1-metre drone orthophotos instead of coarse 30m satellite pixels.

## Slide 15: Limitations, Future Roadmap & Conclusion

🕒 35 Seconds

### WHAT IS ON PPT:

- Current Limitations: 30m satellite resolution requires drone validation; underground utilities currently scoped out.
- Future Roadmap: Municipal BIM integration, PostgreSQL/PostGIS database persistence, CORS GNSS network validation.
- Core Takeaway: BoundaryLens solves 3D vertical property mapping transparently, honestly, and reliably.

Transition: "To conclude our presentation..."

### 🔑 EXACTLY WHAT TO SAY:

"We recognize our current boundaries: 30-metre public elevation cannot replace high-resolution drone survey flights, and underground utilities require municipal utility scans.

Our roadmap includes direct ingestion of municipal BIM models, PostGIS database clustering, and CORS GNSS integration for centimeter-level parcel accuracy.

BoundaryLens proves that India can transition from flat 2D cadastral records to structured 3D vertical property identities — with full legal accountability, zero hallucinated data, and complete evidence provenance.

Thank you, and we are ready for your questions!"

### TERM EXPLAINED SIMPLY:

#### **BIM (Building Information Modeling):**

Digital 3D architectural engineering models that contain exact indoor floor slabs and plumbing.

### KEY POINT TO EMPHASIZE:

End with humility and strength: We know the real-world engineering constraints and have a clear technical path forward.

### **Q: What is your single biggest advantage over existing work?**

A: Provenance and honesty. We fuse multi-source evidence and keep an immutable ledger, ensuring AI assists surveyors rather than fabricating legally hazardous title claims.

## Part 2: Emergency Condensed Pitches

## ⚡ The 2-Minute Elevator Pitch (For Strict Time Limits)

Use this script word-for-word if the judges tell you: "You only have 2 minutes, be quick."

"Respected judges, Indian land records are currently locked in 2D flat maps, but our cities have grown vertically into multi-storey complexes. Under **SIH26011**, **BoundaryLens** solves 3D vertical property delineation without fabricating official government data.

Here is how it works in 3 steps: First, our GIS engine checks whether building footprints sit cleanly inside legal revenue parcels. Second, we extract true ground elevation from bare-earth DEM and estimate floor counts using a Random Forest model trained on 12,700 real Bengaluru labels.

Third, our AI Isolation Forest catches boundary encroachments and routes them to human reviewers. As you see on screen, clicking a building displays its terrain elevation, floor stack, and proposed 3D identifier. When I click 'REJECT', the action is immutably logged in our registrar activity trail.

With 124 passing unit tests and zero fake ULPINs, BoundaryLens gives India a realistic, legally sound pathway from 2D cadastral maps to 3D urban governance. Thank you!"

## 🕒 The 5-Minute Standard Pitch (Recommended Timing Breakdown)

- **0:00 – 0:45 (The Problem):** Introduce Team areonerds; explain that 2D cadastral maps cannot represent vertical property ownership (Slides 1–3).
- **0:45 – 1:30 (The Solution & Pipeline):** Explain the 2D-to-3D hierarchy, DEM terrain sampling, and open data manifests (Slides 4–6).
- **1:30 – 2:30 (Machine Learning & AI):** Explain the 12,700-label Random Forest floor model and Isolation Forest anomaly detection (Slides 7–10).
- **2:30 – 4:00 (Live Interactive Demo):** Switch to localhost:8000. Click a normal building (green), click an anomaly (red), show the proposed 3D linkage, click REJECT, and open the Audit Logs tab (Slide 14).
- **4:00 – 4:30 (Results & Governance):** Mention 124 passing tests, 2,734 buildings, and emphasize that proposed linkages assist human authorities rather than replacing them (Slides 12–13).
- **4:30 – 5:00 (Conclusion & Q&A Hand-off):** Acknowledge limitations (30m satellite vs 1m drone), outline BIM roadmap, and open for judge questions (Slide 15).

## Part 3: Critical Defense Protocols

### 🚫 7 Things You Must NEVER Say to SIH Judges

1. **NEVER say:** "We are officially issuing 3D ULPIN numbers."  
**CORRECT:** "We generate proposed 3D vertical spatial linkages for administrative evaluation and reviewer adjudication. Official legal ULPIN issuance remains with the revenue department."
2. **NEVER say:** "Our model has 99% accuracy in predicting floors."  
**CORRECT:** "Our Random Forest achieves a calibrated Mean Absolute Error of 2.22 floors and 59.8%  $\pm 1$  accuracy on open satellite data. We do not invent unrealistic accuracy metrics."
3. **NEVER say:** "Satellite DEM alone gives us the exact floor count."  
**CORRECT:** "Public 30-metre DEM/DSM provides an elevation signal, but cannot directly measure indoor floor slabs. We treat elevation as one evidence input alongside footprint geometry and supervised ML."
4. **NEVER say:** "AI makes the final decision on boundary disputes."  
**CORRECT:** "AI flags multi-dimensional anomalies, but human officers retain exclusive decision-making authority via our Reviewer Audit Gate."
5. **NEVER say:** "We used high-resolution Google Street View or private government deed databases."  
**CORRECT:** "We strictly use legal, public, documented open datasets (OpenCity, OpenStreetMap, Copernicus, USGS SRTM) with complete signed manifests."
6. **NEVER say:** "Our audit logs are saved permanently on the blockchain."  
**CORRECT:** "In our hackathon prototype, reviewer actions are logged into a structured session ledger; the production roadmap connects this to a persistent PostgreSQL/PostGIS database."
7. **NEVER say:** "We also mapped all underground metro tunnels and sewer lines."  
**CORRECT:** "Underground infrastructure is deliberately scoped out of this MVP due to the absence of public subsurface GIS layers; our spatial entity model is architected to extend downward once utility point clouds are ingested."

### The "Last 30-Minute" Pre-Judging Checklist

| Step | Action Required                         | Expected Verification                                                                                                                                                                                                                                                                                                          |
|------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Check Laptop Power &amp; Display</b> | Plug in AC adapter; set screen sleep to "Never"; connect HDMI/projector duplicate mode.                                                                                                                                                                                                                                        |
| 2    | <b>Activate Virtual Environment</b>     | In PowerShell: <code>cd D:\BoundaryLens; .\env\Scripts\activate</code>                                                                                                                                                                                                                                                         |
| 3    | <b>Run Pre-Demo Health Check</b>        | Run <code>python scripts/demo_health_check.py</code> → Must output [PASS] across all 7 items and <b>DEMO READY</b> .                                                                                                                                                                                                           |
| 4    | <b>Start Demo Server</b>                | Run <code>python run_demo.py</code> (or double click <code>start_demo.bat</code> ) → Server starts on port 8000.                                                                                                                                                                                                               |
| 5    | <b>Open Browser Tab</b>                 | Open Chrome at <code>http://localhost:8000/</code> → Confirm MapLibre 3D canvas loads South Bengaluru.                                                                                                                                                                                                                         |
| 6    | <b>Pre-Select Target Buildings</b>      | <ul style="list-style-type: none"><li>- Normal: Click <code>osm_way_93697573</code> (Green, 100% overlap, 904.2m elevation).</li><li>- Anomaly: Click <code>osm_way_130829155</code> (Red/Amber, anomaly score -0.036).</li><li>- Floors: Click <code>osm_way_347151672</code> (Observed 2 floors, high confidence).</li></ul> |
| 7    | <b>Test Reviewer Action</b>             | Click REJECT → Confirm status updates to <code>REVIEWER_REJECTED</code> → Check Audit Logs tab shows entry.                                                                                                                                                                                                                    |
| 8    | <b>Prepare Offline Backup</b>           | Keep <code>frontend/public/Screenshot_31-8-2026_21530_localhost.jpeg</code> open in background image viewer just in case.                                                                                                                                                                                                      |